

Strategy Backtest Suite Results

Date: 2026-09-12 · Freqtrade research only · No live trading · No hyperopt

Setup

- Pairs: BTC/USDT, ETH/USDT, SOL/USDT, BNB/USDT, XRP/USDT
- Timeframes: 1d, 4h, 1h (5m/15m skipped — no local data)
- Periods A–E tested; Period F skipped (data starts ~2024-07-05)
- Risk scenarios: 2% / 3% / 5% via ATR%-scaled position size
- Total runs: 273

Headline Rankings

Rank	Strategy	Note
Best Overall	Score4WindowStrategy	STRONG on A_1m, B_6m, E_2y
Best Short Term	Score4WindowStrategy	A_1m Ret~3.36% PF~5.4
Best Medium Term	Score4WindowStrategy	B_6m Ret~2.49% PF~1.79
Best Long Term	Score4WindowStrategy	E_2y Ret~6.81% PF~1.42
Robust #2-5	RSI/MACD/TSMom/Donchian	2 active periods each
C_1y window	Almost all FAIL	Hostile regime for defaults

Status Distribution

Status	Count
FAIL	177
ACTIVE_CANDIDATE	63
WATCH	17
LOW_ACTIVITY	13
STRONG_CANDIDATE	3

Strategy Summary (top)

Strategy	Status	AvgReturn_1d_r3	AvgMaxDD_1d_r3	AvgPF_1d_r3	AvgSharpe_1d_r3	BestTimeframe
Score4WindowStrategy	STRONG_CANDIDATE	2.5499	2.4791	2.1248	2.113	1d
SeriesMomentum	ACTIVE_CANDIDATE	0.2167	0.5537	0.997	2.0457	1d
ScoreSeriesMomentum	ACTIVE_CANDIDATE	0.2397	0.6098	1.1464	1.5348	1d
APPullback	ACTIVE_CANDIDATE	0.2694	0.5007	5.064	1.184	1d
VolatilityBreak	ACTIVE_CANDIDATE	0.1748	0.6264	2.1593	1.0061	1d
IndicatorConf	ACTIVE_CANDIDATE	-0.0514	0.7269	2.338	0.9482	1d
CDMomentum	ACTIVE_CANDIDATE	0.3458	0.5436	2.8205	0.9228	1d
DonchianChannel	ACTIVE_CANDIDATE	0.0802	0.6838	2.2297	0.8785	1d
DonchianBreakout	ACTIVE_CANDIDATE	0.1778	0.6401	1.7833	0.7385	1d
EnglerBreakout	ACTIVE_CANDIDATE	0.1564	0.5753	1.784	0.7127	1d
TimeConfirmedBreakout	ACTIVE_CANDIDATE	-0.0343	0.7032	1.6648	0.6361	1d
UptrendStrategy	ACTIVE_CANDIDATE	0.1966	0.4927	1.5545	-19.8401	1d

Full Analysis Notes

Strategy Backtest Suite — Final Analysis

Date: 2026-09-12
Scope: Research backtests only (no live trading, no hyperopt)
Engine: Freqtrade backtesting
Pairs: BTC/USDT, ETH/USDT, SOL/USDT, BNB/USDT, XRP/USDT
Exchange data: Binance via cloudmirror (`data-api.binance.vision`)
Config: `user\_data/config\_suite\_binance5\_cloudmirror.json`

Decisions (where config was incomplete)

Topic	Decision
Pair universe	5 liquid USDT pairs (matches prior baseline harness; full 100-pair dry-run list too slow for 273-job mat
Timeframes	<code>1d / 4h / 1h</code> tested. <code>5m/15m</code> skipped (not present in local data; full history download too large for
Period F	(2024-03-12→2026-09-12) SKIPPED – local OHLCV starts ~2024-07-05
Risk	2% / 3% / 5% Modeled as <code>size_pct = risk_pct / atr_pct</code> (cap 25%). Score4Window uses <code>score * risk_pct / atr_pct</code>
Higher risk	≠ better Rankings use Sharpe / PF / expectancy / DD – not raw return alone
Hyperopt	Not run (per brief)

Artifacts

- `reports/strategy\_backtest\_results.csv` (273 runs)
- `reports/strategy\_summary.csv`
- `reports/strategy\_suite\_report.md` / `.json`
- Copies at repo root: `strategy\_backtest\_results.csv`, `strategy\_summary.csv`

Lookahead

`freqtrade lookahead-analysis` on Score4Window, DonchianBreakout, RSIMomentum, TimeSeriesMomentum, MACDMomentum (`1d`, timerange covering recent year):
No bias detected (0 biased signals / exits).

Headline rankings

- BEST OVERALL (robust):** Score4WindowStrategy — only strategy with **3 STRONG** period tags (A_1m, B_6m, E_2y)
- BEST SHORT TERM (A_1m):** Score4WindowStrategy — Ret≈3.36%, PF≈5.4, Sharpe≈8.6, 19 trades
- BEST MEDIUM TERM (B_6m):** Score4WindowStrategy — Ret≈2.49%, PF≈1.79, Sharpe≈1.08, 42 trades
- BEST LONG TERM (E_2y):** Score4WindowStrategy — Ret≈6.81%, PF≈1.42, Sharpe≈0.82, 203 trades
- MOST ACTIVE QUALITY:** Score4WindowStrategy (highest trade count among quality-tagged runs)
- LOWEST DRAWDOWN (among non-fail multi-period):** VWAPPullback tends to keep DD lower than Score4Window on C_1y, but expectancy still negative there
- BEST PROFIT FACTOR (short windows):** Score4Window A_1m (PF≈5.4); over full matrix many alts inflate PF with few trades
- BEST RISK/REWARD:** Score4Window on A_1m / B_6m (Sortino strong); alts do not beat it consistently
- BEST OOS / ROBUSTNESS:** Score4Window (3 active periods). Next: RSIMomentum, MACDMomentum, TimeSeriesMomentum, DonchianBreakout, ATRVolatilityBreakout, BollingerBreakout (2 periods each)
- SCORE4WINDOW VS BEST ALT:** On `C_1y` (2025-09→2026-09) almost everything **FAIL**s (negative expectancy), including Score4Window. No alternative earns a clean pass on that window with default rules. Prefer multi-period robustness over C_1y-only ROI.

Classification summary (all 273 runs)

Status	Count
FAIL	177
ACTIVE_CANDIDATE	63
WATCH	17

| LOW_ACTIVITY | 13 |
| STRONG_CANDIDATE | 3 (all Score4Window) |

Important regime note

C_1y and **D_13m** are hostile for default-parameter trend/breakout rules on this 5-pair set. Strategies that look fine on **B_6m** / **E_2y** often collapse on **C_1y** → treat single-window winners as **OVERFIT_RISK** unless they also pass other periods.

Risk sweep (C_1y, 1d)

Raising risk 2→3→5 mainly **scales drawdown**; it does **not** rescue negative-expectancy systems. Do not prefer 5% risk on return alone.

Timeframe sweep

On this matrix, **1d >> 4h/1h** for quality. Lower TFs produced more trades but worse PF/Sharpe / deeper DD for most candidates (including Score4Window).

Candidate shortlist for later (controlled) hyperopt

Only after OOS holdout design:

1. Score4WindowStrategy (baseline champion)
2. RSIMomentum
3. MACDMomentum
4. TimeSeriesMomentum
5. DonchianBreakout
6. ATRVolatilityBreakout
7. BollingerBreakout
8. VWAPPullback (activity/DD interesting, verify expectancy)

Not recommended now: Supertrend (weak / period-fragile after fix), pure mean-reversion set (RSI/ZScore/Bollinger MR), TTMSqueeze defaults, TrendPullbackRetest defaults.

Supertrend note

Initial Supertrend helper had ATR-warmup NaN poisoning (0 trades). Fixed in `strategy\_common.supertrend`; suite re-ran Supertrend jobs. Still not a top robustness pick.

Full trade-level metrics: see strategy_suite_results.xlsx (Backtest Results sheet).
Lookahead: no bias detected on Score4Window + momentum/breakout shortlist.